

Renke Wang

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Professional Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in model predictive control, optimal control, robust control, mathematical optimization, and scheduling-control co-design for constrained, networked, nonlinear, and multi-agent dynamical systems. Develops optimization-based control, real-time scheduling, and decision-making methods that coordinate control inputs, resource allocation, sequencing, and timing under dynamical and shared-resource constraints. Experience includes tube-based MPC, nonlinear and nonconvex optimization, mixed-integer optimization, distributed consensus optimization, Lyapunov-based stability analysis, robust control under timing uncertainty, event-driven simulation, computational benchmarking, and runtime and scalability analysis.

Education

George Mason University

Ph.D. Candidate in Electrical and Computer Engineering

Expected 2027

Dalian University of Technology

Master of Engineering, Control Science and Engineering

Dalian University of Technology

Bachelor of Engineering, Automation

Technical Skills

Model Predictive, Optimal, and Robust Control: Model Predictive Control (MPC); constrained optimal control and receding-horizon optimization; tube-based robust MPC; state-space modeling and feedback control; state and input constraints; piecewise-constant control parameterization; dynamic state-feedback gains; scheduling-control co-design; robust control under timing uncertainty; closed-loop simulation and performance evaluation.

Control Theory and Stability Analysis: Lyapunov-based stability analysis; Linear Matrix Inequality (LMI)-based controller synthesis; asymptotic stability; exponential stability; stochastic stability; state-feedback controller design; positive systems; switched systems; Markov Jump Linear Systems (MJLS); dwell-time methods; event-triggered control; L1-gain analysis and controller synthesis.

Mathematical Optimization and Computational Methods: Linear Programming (LP); Mixed-Integer Programming (MIP); Mixed-Integer Linear Programming (MILP); Nonlinear Programming (NLP); Mixed-Integer Nonlinear Programming (MINLP); nonlinear and nonconvex constrained optimization; centralized and distributed optimization; Alternating Direction Method of Multipliers (ADMM); Consensus Alternating Direction Method of Multipliers (C-ADMM); numerical optimization workflows; solver diagnostics; sensitivity analysis; runtime analysis; scalability analysis.

Scheduling, Planning, and Control Co-Design: Resource-constrained, real-time, and priority-based scheduling; resource allocation; task allocation; task and resource-occupation sequencing; priority assignment and dispatching; joint scheduling and control; routing and shortest-path planning; joint path selection and scheduling; decision-tree search; graph-search methods; Dijkstra's algorithm; A* search; multi-step lookahead; online decision-making and multi-agent coordination under shared and heterogeneous resource constraints.

Modeling, Simulation, and Real-Time Systems: Dynamical-system modeling; differential-drive mobile robot modeling; networked and cyber-physical systems; resource-constrained and event-driven control; event-driven simulation; Monte Carlo simulation; trajectory and system-performance evaluation; model validation; baseline benchmarking; parallel local optimization workflows; automated-vehicle, robotic, and multi-agent simulation.

Learning-Based Sequential Decision-Making: Supervised neural policy learning from exact optimization and decision-tree search; Deep Reinforcement Learning (DRL); Proximal Policy Optimization (PPO); actor-critic methods; event-driven Markov Decision Processes (MDPs); learning-based path selection; priority assignment; shared-resource scheduling.

Tools and Technologies

- **Programming and Data Analysis:** Python, SQL, MATLAB, Pandas, NumPy, SciPy, Matplotlib, and Microsoft Excel.
- **Optimization, Control, and Machine Learning:** Gurobi, CasADi, YALMIP, MATLAB Optimization Toolbox, MATLAB Control System Toolbox, and PyTorch.
- **Development and Computing:** Git, GitHub, Linux, Jupyter Notebook, and LaTeX.

Research Experience

Research Assistant at George Mason University

*Fairfax, VA
August 2022 – Present*

Optimization-Based Scheduling and Control Co-Design for Shared-Resource Systems

- Developed Contention-Resolving Model Predictive Control (CR-MPC) formulations that jointly optimize continuous control inputs, priority and sequencing decisions, resource access, and occupation timing for dynamical systems under state, input, timing, and shared-resource constraints; implemented finite-horizon MPC subproblems in CasADi with piecewise-constant control inputs and evaluated the framework for automated vehicles, warehouse merging zones, and multi-intersection traffic networks.
- Developed a control-aware scheduling framework for multiple lighter-than-air agents sharing limited heterogeneous computational resources for remote wind estimation; modeled processor-dependent execution times and scheduling-induced nonlinear task timing, optimized resource assignment and access sequencing using decision-tree search, and enforced delay bounds under wind disturbances to preserve trajectory feasibility.
- Developed the real-time scheduling component of an optimization-based recommendation system for allocating limited human attention among multiple robots requiring time-critical obstacle avoidance; modeled operator interventions as deadline-constrained tasks and optimized which robot should receive attention and when using decision-tree search.

Robust Model Predictive Control under Distribution-Free Timing Perturbations

- Developed a tube-based robust MPC framework for Linear Time-Invariant (LTI) systems subject to distribution-free control-update timing perturbations; designed timing-dependent dynamic feedback gains, derived the resulting closed-loop error dynamics, and established asymptotic stability and the existence of a forward invariant set.
- Developing Koopman operator-based lifted linear models of nonlinear system dynamics through data-driven identification of Koopman operators from trajectory data, with applications to Model Predictive Control (MPC) and robustness analysis under timing perturbations.

Distributed Scheduling and Control Co-Design for Networked Control Systems

- Developed a C-ADMM-based distributed optimization framework that decomposes multi-layer scheduling, sequencing, resource-allocation, and timing decisions into resource-specific subproblems while coordinating coupled task timing through consensus constraints.
- Implemented parallel local optimization workflows and benchmarked closed-loop cost, solution quality, computation time, and scalability against advanced optimization-based methods and baseline dispatching strategies.

Learning-Based Path Selection and Shared-Resource Decision-Making

- Developed a supervised neural scheduling policy trained on optimal decisions generated by exact CR-MPC decision-tree search, using optimization-generated decisions to learn event-driven contention-resolution policies under shared-resource constraints.
- Developing an event-driven PPO actor-critic framework for joint path selection and shared-resource scheduling, extending fixed-route scheduling to online routing, priority assignment, and sequential decision-making in multi-agent systems.

Graduate Researcher at Dalian University of Technology

*Dalian, China
Sept 2017 – June 2020*

Stability Analysis and Controller Design for Positive and Switched Systems

- Derived Lyapunov-based stability conditions for positive switched systems and Markov Jump Linear Systems (MJLS) under dwell-time constraints, stochastic mode transitions, and fully or partially known transition rates.
- Designed state-feedback and L1 controllers using piecewise-linear copositive Lyapunov functions, establishing stochastic and exponential stability and L1-gain performance under external disturbances and controller failures.

Selected Technical Implementations

- **Differential-Drive Mobile Robot Control:** Implemented closed-loop simulations for a differential-drive mobile robot using unicycle kinematics and polar error coordinates; evaluated trajectory convergence across multiple initial positions and headings.
- **Nonlinear Vehicle Scheduling at Traffic Intersections:** Implemented a fixed-precedence nonlinear vehicle-scheduling formulation coupling intersection entry and exit timing with piecewise-constant acceleration, vehicle-motion constraints, conflict-region occupancy, and rear-end safety constraints; evaluated four-vehicle intersection scenarios through numerical simulation.

Selected Publications and Manuscripts

- Renke Wang and Ningshi Yao. *Distributed Scheduling and Control Co-Design for Networked Control Systems with Multi-Layered Shared Resources*. IEEE Transactions on Control of Network Systems (TCNS), under review.
- Renke Wang, Mengxue Hou, and Ningshi Yao. *Robust Co-Design of Scheduling and Control for Lighter-than-Air Agents with Heterogeneous Resources*. International Journal of Robust and Nonlinear Control, under review.
- J. Lian and R. Wang. *Stochastic Stability of Positive Markov Jump Linear Systems with Fixed Dwell Time*. Nonlinear Analysis: Hybrid Systems, 40 (2021): 101014.
- R. Wang and N. Yao. *Contention-Resolving Model Predictive Control for Coordinating Automated Vehicles at a Multi-Lane Traffic Intersection*. 2025 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2025.
- Rajul Kumar, Renke Wang, and Ningshi Yao. *One Human and Multi-Robot Collaboration: Evaluating the Impact of Attention-Guided On-Screen Recommendations on Human Multitasking Efficiency, Cognitive Load, and Trust*. 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), IEEE, 2025.
- R. Wang and N. Yao. *Robust Model Predictive Control for Networked Control Systems with Timing Perturbations*. 2024 American Control Conference (ACC). IEEE, 2024.
- N. Yao and R. Wang. *Optimal Real-Time Human Attention Allocation and Scheduling in a Multi-human and Multi-robot Collaborative System*. 2024 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2024.

Teaching Experience

Teaching Assistant, George Mason University

Fairfax, VA

Aug 2022 – Dec 2022

- Delivered lectures and led or co-led laboratory sessions for Continuous-Time Signals and Systems; explained technical concepts, guided laboratory work, graded assignments, and supported undergraduate students.
- Graded coursework and answered student questions for Classical Systems and Control Theory, providing clarification on course concepts and assignments.

Honors and Professional Service

- Summer Research Assistantship Award, George Mason University Graduate Division, 2026.
- Reviewer for IEEE Transactions on Control Systems Technology; IEEE Transactions on Systems, Man, and Cybernetics: Systems; and IEEE conferences including ACC, CDC, and CCTA, 2023–Present.